

2(a)	any three from: 1 open in, day / light ; 2 to take in carbon dioxide for, photosynthesis / light-independent stage / Calvin cycle ; 3 close in, night / dark ; 4 to decrease water loss by transpiration ;	3
2(b)	any six from: 1 hydrogen ions / protons / H ions / H⁺, leave (cell) using, energy / ATP ; 2 low H⁺ (in cell) / <u>more</u> negative charge ; 3 K⁺ / potassium ion, channel (proteins) open OR K⁺, move into / enter, cell (by facilitated diffusion) ; 4 Cl⁻ ions, move into / enter (cell) ; 5 water potential of cell decreases ; 6 water moves in (to cell) by <u>osmosis</u> ; 7 (cells) become turgid / swell / increase in volume / expand ; 8 (guard cell) inner wall thicker / outer wall thinner, so cells, bend / curve ;	6
2(c)	1 (environment) light needed, to keep cycle normal / to achieve same peak % of stomata opening ; 2 genes have a role as (regular), rhythm / pattern / cycle, continues, in darkness / without light ;	2